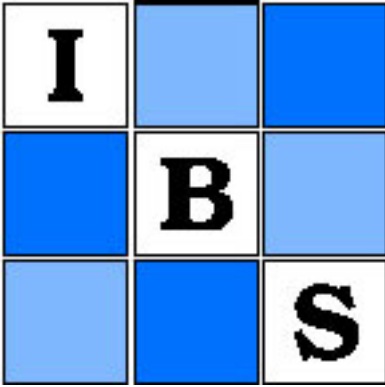




Generalized Estimating Equations for Ordinal Data: A Note on Working Correlation Structures

Author(s): Thomas Lumley

Source: *Biometrics*, Vol. 52, No. 1 (Mar., 1996), pp. 354-361

Published by: [International Biometric Society](#)

Stable URL: <http://www.jstor.org/stable/2533173>

Accessed: 30/03/2014 17:04

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <http://www.jstor.org/page/info/about/policies/terms.jsp>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.



International Biometric Society is collaborating with JSTOR to digitize, preserve and extend access to *Biometrics*.

<http://www.jstor.org>

Generalized Estimating Equations for Ordinal Data: A Note on Working Correlation Structures

Thomas Lumley

NHMRC Clinical Trials Centre, University of Sydney,
New South Wales 2006, Australia

SUMMARY

Generalized estimating equations (Liang, K. Y. and Zeger, S., 1986, *Biometrika* **73**, 13–22) allow longitudinal or clustered data to be modeled with minimal assumptions about their dependence structures. Association structures for polytomous data have generally required the estimation of a large number of parameters. In many applications involving repeated categorical data, an ordinal structure is present. A range of association structures and computational methods for ordinal categorical data is described, based on the cumulative odds ratio, which allows much more parsimonious models. This permits the generalized estimating equation methodology to be used for smaller sets of ordinal data and with less effort expended on modeling associations. The method is illustrated on sets of ordinal data from medical studies.

1. Introduction

Generalized estimating equation (GEE) models require the user to specify a working correlation structure. For binary and continuous data, a high level of efficiency can be attained with comparatively simple correlation structures; however, there is a shortage of simple correlation structures for polytomous data. One obvious possibility is a saturated model with a separate parameter for each of the possible pairs of responses ($(n_t n_c - 1)^2/2$ for n_t times and n_c categories). Miller, Davis, and Landis (1994), who extended GEE models to polytomous data, went further and have a polysaturated model with a separate complete set of correlations for each combination of explanatory variables, although this model is presented at least in part to illustrate its relationship to the weighted least-squares methodology of Koch et al. (1977), which turns out to be a one-step version of the iterative method for this GEE model. There are a lot of conjectures that efficiency could be improved by “careful modeling of the correlation structure.” This seems to be missing the point in many cases. Careful modeling of variance structures is difficult and often unprofitable even when they are of primary interest. The sensitivity of variance component models is notorious, even in the analysis of variance situation. It seems even less worthwhile to devote time and information to modeling covariance structures in GEEs, where they are only nuisance parameters. According to the simulations and examples in papers such as those by Liang and Zeger (1986) and Liang, Zeger, and Qaqish (1992), the asymptotic gain in efficiency for the marginal regression parameters over independence models is about 5–10%, most of which can be achieved with far simpler correlation structures. For moderate sample sizes, the loss of efficiency and accuracy from a model with too many variance parameters may be large, and there is certainly an increase in the rate of numerical singularities in the fitting process. In particular, if the highest or lowest category in an ordinal regression has comparatively few counts, the fitting process can easily break down, with the correlations and cutpoint corresponding to that category dominating the calculations.

For these reasons, it is useful to have working correlation structures for polytomous data that produce reliably positive-definite correlation matrices and require few parameters and little data. The original GEE paper (Liang and Zeger, 1986) gave several such structures, none of which is suitable without modification for polytomous data. The major difficulty is that multinomial

Key words: Dependent data; Estimating equations; Cumulative link models; Odds ratio; Ordinal data.

data require negative as well as positive correlations, and it is not obvious how to model this structure parsimoniously. Lipsitz, Kim, and Zhao (1994) suggested a method that constrains the correlations at different times between two categories: for example, an exchangeable model in which the correlation between two given categories is the same at all pairs of times $t \neq t'$. This method gives a substantial reduction in the number of parameters over the methods of Miller et al. (1994), but still requires a minimum of $n_c(n_c - 1)/2$ parameters for n_c categories. For five categories, not an unreasonable number on an ordinal scale, this is 10 parameters for even the most restricted correlation structure.

Another suggestion for modeling multinomial data, described by Gange et al. (1995) involves replacing the block-diagonal covariance matrix $\mathbf{A}$ from the independence model with a matrix $\mathbf{A}^{1/2}\mathbf{R}\mathbf{A}^{1/2}$, where $\mathbf{R}$ can be modeled parsimoniously in a variety of ways by setting most of its components to zero. The authors stated, incorrectly, that $\mathbf{R}$ is the correlation matrix. In fact, modeling the correlation matrix in a similar way, by setting most between-time correlations to zero does not provide useful models, as it is difficult to ensure that the resulting matrix is positive definite. The models of Gange et al. (1995) would be very useful for unordered categorical data but, as that paper admits, for ordered data it is better to have a model that takes into account the structure of the data.

2. Models for Ordinal Data

While models for ordinal data use the multinomial likelihood to describe the random variation, the systematic part of the model should take advantage of the ordinal structure of the data. This applies not only to the marginal models but to the correlation between observations, implying that we should look for a measure of association between two ordinal variables that allows straightforward calculation of the probabilities for each cell and, thus, the required correlation matrix.

The need to use the ordinal structure of data in modeling is emphasized by Agresti (1990). One recommended class of models, the uniform association models, describes the association in a table by a single odds ratio. For models using scores this is the local odds ratio between groups of four adjacent cells; for cumulative logit models, it is the cumulative odds ratio. Either of these would provide a simple one-parameter description of the association between responses at two different times. It would be more consistent to use the local odds ratio for a parallel odds regression and the cumulative odds ratio for a proportional odds regression, but the distinction is unlikely to be important. In this note, we deal with only the cumulative odds ratio and McCullagh's (1980) proportional odds regression model. Gange et al. (1995) in a discussion of methods for ordinal data in ophthalmology, suggested that the global odds ratio parametrization would be most appropriate for ordinal data and the theory of these methods has been developed in detail by Heagarty and Zeger (1994), although their algorithms for estimating the odds ratio differ from those presented here. A likelihood-based model for bivariate ordinal data was constructed along these lines by Dale (1986). For bivariate data, the joint distribution is then a discretization of a bivariate Plackett distribution (Plackett, 1965) and maximum likelihood estimators of parameters in the global odds ratio and the marginal models are uncorrelated. This orthogonality should lead to high efficiency for the GEE approach, even though the odds ratio is not being estimated by maximum likelihood. Molenberghs and Lesaffre (1994) constructed a full likelihood-based model, the multivariate Dale model, for repeated ordinal measurements by generalizing the Plackett distribution to multivariate data. While this has the well-known advantages of likelihood-based analysis, it appears substantially more computationally intensive and requires the number of measurements on each cluster to be the same. Ezzet and Whitehead (1991) presented a model for ordinal cross-over trials in which the response is considered as a discretization of a latent variable that is the sum of a logistic and a normal variable. This gives an ordinal random-effects model, suitable in principle for other forms of exchangeable ordinal data. It lacks the robustness of the GEE methodology to errors in the assumed correlation structure, and the fitting process, which requires numerical integration, may only be practical for small sets of discrete covariates.

Given the choice of the common odds ratio to describe association between two fixed times t and t' , it is necessary to specify how it is assumed to vary with (t, t') . One possibility is to estimate it separately for each (t, t') , which still gives about $(n_c - 1)^2$ times fewer parameters than a saturated model and guarantees a positive (semi)definite correlation matrix. More parsimonious structures include an exchangeable model where it is constant for all pairs of times, a one-dependent model with odds ratio different from 1 only for adjacent times or an autoregressive-type model where the odds ratio is of the form $\tau^{1/(|t-t'|)}$.

3. Computation

There are two main computational issues: estimating the common odds ratio from the data and estimating the covariance matrix from the common odds ratio.

First, we establish some notation. At each time $t = 1, \dots, n_t$ and for each group ($k = 1, \dots, K$), we have an ordinal categorical response $R_{kt} \in \{1, \dots, n_c\}$, where the dependence on k will usually be suppressed. To fit a GEE, we convert this into $n_c - 1$ binary responses $Y_{tc} = I(R_t = c)$, omitting category n_c for identifiability. The expected values of these responses under the model are μ_{tc} . The cumulative probabilities, used to construct the cumulative odds ratio, are $\gamma_{tc} = \Pr(R_t \leq c)$.

Estimating the covariance matrix from the common odds ratio is straightforward. The covariance of the binary variables Y_{tc} and $Y_{t'c'}$ is given by

$$\text{cov}(Y_{tc}, Y_{t'c'}) = E(Y_{tc}Y_{t'c'}) - E(Y_{tc})E(Y_{t'c'}).$$

The second term is the product of the predicted means μ_{tc} and $\mu_{t'c'}$ and the first is the joint probability $\mu_{(cc')} = \Pr(R_t = c \cap R_{t'} = c')$. This can be obtained by differencing the joint cumulative probabilities $\gamma_{tct'c'} = \Pr(R_t \leq c \cap R_{t'} \leq c')$:

$$\Pr(R_t = c \cap R_{t'} = c') = \gamma_{t(c+1)t'(c'+1)} - \gamma_{tct'(c'+1)} - \gamma_{t(c+1)t'c'} + \gamma_{tct'c'}.$$

The relationship between the joint and marginal cumulative probabilities and the odds ratio gives Mardia's (1970) formula for the joint cumulative probability

$$\gamma_{tct'c'} = \frac{f - \sqrt{f^2 - 4\tau(\tau - 1)\gamma_{tc}\gamma_{t'c'}}}{2(\tau - 1)},$$

where $f = 1 - (1 - \tau)(\gamma_{tc} + \gamma_{t'c'})$

Dale (1986), who described bivariate ordinal models based on the odds ratio, used maximum likelihood methods to estimate this "global cross-ratio." These would be useful if the association were to be estimated as a function of the covariates, but for these simple models they are not necessary. The method presented here, while admittedly *ad hoc* in construction, requires rather less computation and programming than a quasi-Newton scheme and is more numerically stable for sparse data sets.

There are various methods of estimating the common odds ratio from a table. Maximum likelihood is possible but computationally intensive. A simpler estimate is obtained by considering every possible double dichotomy of the two variables to give a 2×2 table. If a table has counts m_{ij} in each cell and is partitioned at (c, c') , the resulting 2×2 table has counts

$$\begin{aligned} A_{cc'} &= \sum_{i>c, j>c'} m_{ij}, \\ B_{cc'} &= \sum_{i\leq c, j>c'} m_{ij}, \\ C_{cc'} &= \sum_{i>c, j\leq c'} m_{ij}, \\ D_{cc'} &= \sum_{i\leq c, j\leq c'} m_{ij}. \end{aligned}$$

The odds ratio is

$$\tau_{(cc')} = \frac{A_{cc'}D_{cc'}}{B_{cc'}C_{cc'}},$$

and the approximate variance of $\log \tau_{cc'}$ is

$$V_{(cc')} = \frac{1}{A_{cc'}} + \frac{1}{B_{cc'}} + \frac{1}{C_{cc'}} + \frac{1}{D_{cc'}}.$$

While the odds ratio estimates are not independent they are each consistent, and the weighted mean of the log odds ratios with weights inversely proportional to the variances will be consistent and reasonably efficient. A minor reduction in bias and increase in stability can be obtained by adding 0.5 to each of A , B , C , and D .

The principal difficulty in constructing an odds ratio estimate for the GEE is that there is only one observation in each of the tables of $R_{kt} \times R_{kt'}$ for the k th individual at times t and t' . Some

way to “borrow strength across groups” is needed. One solution is to estimate the odds ratio from that in the bivariate marginal table of $R_t \times R_{t'}$. This crude odds ratio estimate τ_0 is, in general, not the same as the stratified odds ratio that we require. In most repeated-measures data, it will be biased upward, but the bias can be estimated and corrected. Given an estimate of the common (stratified) odds ratio, we calculate the expected value of the crude marginal odds ratio and, thus, adjust the estimate up or down to match the actual crude odds ratio. If we have a current estimate $\hat{\tau}$ of the (stratified) odds ratio, we can calculate the joint cumulative probabilities and the joint probabilities $\mu_{(ij)k}$ for cell (i, j) in the k th subject as described previously. These joint probabilities are then added:

$$\hat{m}_{ij} = \sum_k \mu_{(ij)k}$$

to give an estimate $\hat{m}_{ij}$ of the count in the (i, j) cell of the bivariate marginal table of $R_t \times R_{t'}$ under the assumption that the odds ratio is $\hat{\tau}$. From this estimated table, we calculate an estimated crude odds ratio τ'_0 . A reasonable estimate for the ratio τ/τ_0 of the stratified odds ratio to the crude odds ratio is $\hat{\tau}/\tau'_0$, its value if $\hat{\tau}$ were the true stratified odds ratio. This gives a new estimate,

$$\hat{\tau}_{new} = \frac{\hat{\tau}}{\tau'_0} \tau_0,$$

and the procedure can be iterated to get more accuracy. Convergence is usually rapid. The limiting value is consistent for the true global odds ratio, and estimates after a fixed number of iterations are at least $K^{1/2}$ -consistent for their expected values, all that is required by the GEE methodology.

As usual in GEE estimation, we would alternate between iterations for τ and the marginal regression parameters. In the usual repeated data situation, where $\tau_0 \geq \tau \geq 1$, the estimate will stay in the interval $[1, \tau_0]$ provided that the initial estimate is in this interval. The degenerate estimates that sometimes occur with correlation-based estimates are thus rarer with this method.

This gives a model with one parameter for each pair of time points. These can be combined to give more parsimonious models. For example, an exchangeable model has only one odds ratio, which would be calculated as the geometric mean of the estimates $\tau_{tt'}$ (a weighted mean would be possible but is unlikely to be worthwhile, as the variance of $\hat{\tau}$ will usually be similar for each pair of times). A nonstationary m -dependence model sets $\tau_{tt'}$ to 1 if t and t' are separated by more than m points. A stationary m -dependence model odds ratio has one parameter τ_l for all pairs (t, t') with $|t - t'| = l \leq m$, and again this can be calculated as the geometric mean of the separate estimates $\tau_{tt'}$. Alternatively, especially if the number of observations in each group varies considerably, it may be desirable to add the marginal tables for which the odds ratio will be the same and compute a single odds ratio. The procedure for doing this is the same as that described previously; if two tables have the same odds ratio then they can be combined and the stratified odds ratio, calculated from the crude odds ratio. This is especially useful in analyzing clustered data, where there is no natural “time” ordering and the number of observations in each group can vary considerably. The only disadvantage of this method (apart from some administrative details in computation) is that estimates of all $n_t(n_t - 1)/2$ odds ratios are not available for diagnostic purposes and there is an advantage in stability provided by the larger counts.

4. Examples

This method is demonstrated on two sets of data. One is from a paper by Ezzet and Whitehead (1991) describing an ordinal random-effects model for cross-over trials. The second is a smaller data set from a surgical randomized trial, which provided the original motivation for developing the algorithms presented here.

4.1 Asthma Cross-Over Trial Data

These data come from the paper of Ezzet and Whitehead (1991) on ordinal data in cross-over trials and relate to the comprehension of two information leaflets for inhalation devices used for asthma. The difficulty of comprehension is scored as “Easy,” “Only clear after rereading,” “Not very clear,” or “Confusing.” This data set allows the comparison of maximum likelihood and estimating equation approaches to a random-effects (exchangeable) association structure. The independence and exchangeable working models were used to get the results in Table 1. The study indicates that there is a substantial difference in readability between the two leaflets with the odds of their leaflet A being classified as easier than a given level being about 3.5 times the odds for leaflet B. In addi-

tion, there is a fairly large period effect (odds ratio = 0.3), with each leaflet being easier to understand if presented second. There is no large interaction between leaflet and period. These parameter estimates are similar to those of Ezzet and Whitehead (1991) but with substantially smaller standard errors, around half of theirs. This presumably indicates that the data do not fit the logistic-normal random effects model particularly well. These standard errors are similar to those mentioned by Senn (1993), who presented a very simple reanalysis of the data, ignoring possible interaction/carryover effects and working with a “change” score. The ordinal GEE approach thus confirms Senn’s results with a more reliable method.

Table 1

Ordinal cross-over trial of two information leaflets: parameter estimates and standard errors

	Independence	Exchangeable
Leaflet	1.244 (0.258)	1.250 (0.259)
Period	−1.184 (0.159)	−1.188 (0.159)
Interaction	0.222 (0.195)	0.226 (0.195)

It may also be of interest to compare the association structure for Ezzet and Whitehead’s model and the GEE model. The random-effects model used a normal random effect with estimated variance 1.404. Conditional on the value of the random effect variable, the latent variable had a logistic distribution that has variance $\pi^2/3$. This gives an intrasubject correlation of 0.299. The GEE approach gave an association odds ratio of 2.51. Mardia (1970) gave various estimates for the correlation in a Plackett distribution and stated that

$$\rho_U = \frac{\tau + 1}{\tau - 1} - \frac{2\tau}{(\tau - 1)^2} \log \tau$$

is exact for uniform margins and the best of the simple estimates for normal margins. It should thus perform reasonably well for logistic margins. This formula also gives a correlation of 0.299 as being equivalent to the odds ratio of 2.51. Although this high a level of agreement is presumably a coincidence, it does indicate that the *ad hoc* cumulative odds ratio estimate is a reasonable measure of the within-group association.

4.2 Surgical Randomized Trial Data

These data are from Jorgensen et al. (1995), a randomized trial of abdominal suction to reduce shoulder tip pain after laparoscopic surgery. Inflation of the abdomen during surgery separates the liver from the diaphragm, where it is usually supported largely by surface tension, and places strain on the attachments that connect the two. This strain is felt as referred pain in the shoulder. Under this hypothesis, suction to remove residual gas may reduce shoulder pain but would have only a small effect on general abdominal pain and little or no effect on wound pain. Patients were asked to rate their wound, abdominal, and shoulder pain levels on a visual analog scale morning and afternoon for 3 days after the operation. As there were many zero or very low responses, especially after the first few days, the scale was recoded into five ordered categories. The analysis presented by Jorgensen et al. (1995; Table 2) used the independence working assumption to fit a marginal proportional odds model (without including time after treatment as a covariate). An attempt was made to fit the model with a completely saturated correlation matrix, but the algorithm diverged even when started at the final estimates from the independence model.

The independence, time-saturated, and exchangeable models were fitted to the data on shoulder tip pain, giving the results in Table 3.

The odds ratio for the exchangeable structure was 7.2; those for the time-saturated structure are given in Table 4. In attempting to fit a saturated correlation model, or even one of the category-saturated models of Lipsitz et al. (1994), difficulties arise with pairs of widely separated categories. For example, most patients had low pain levels in the last few days, making it difficult to estimate the correlations between high levels on these days. Although this has little effect on the accuracy of prediction, it does cause the category-specific intercepts to be very unstable. This led to the divergence of the algorithm for a saturated correlation structure. The cumulative odds ratio struc-

Table 2
Data on shoulder tip pain after laparoscopic cholecystectomy (Jorgensen et al., 1995)

I.D. No.	Active treatment	Sex	Age (years)	Pain scores (1 = low, 5 = high)					
				Time 1	Time 2	Time 3	Time 4	Time 5	Time 6
101	Y	F	64	1	1	1	1	1	1
102	Y	M	41	3	2	1	1	1	1
103	Y	F	77	3	2	2	2	1	1
104	Y	F	54	1	1	1	1	1	1
105	Y	F	66	1	1	1	1	1	1
106	Y	M	56	1	2	1	1	1	1
107	Y	M	81	1	3	2	1	1	1
108	Y	F	24	2	2	1	1	1	1
109	Y	F	56	1	1	1	1	1	1
110	Y	F	29	3	1	1	1	1	1
111	Y	M	65	1	1	1	1	1	1
112	Y	F	68	2	1	1	1	1	2
113	Y	M	77	1	2	2	2	2	2
114	Y	M	35	3	1	1	1	3	3
115	Y	M	66	2	1	1	1	1	1
116	Y	F	70	1	1	1	1	1	1
117	Y	M	79	1	1	1	1	1	1
118	Y	F	65	2	1	1	1	1	1
119	Y	F	61	4	4	2	4	2	2
120	Y	F	67	4	4	4	2	1	1
121	Y	F	32	1	1	1	2	1	1
122	Y	F	33	1	1	1	2	1	2
201	N	F	20	5	2	3	5	5	4
202	N	F	50	1	5	3	4	5	3
203	N	F	40	4	4	4	4	1	1
204	N	M	54	4	4	4	4	4	3
205	N	M	34	2	3	4	3	3	2
206	N	F	34	3	4	3	3	3	2
207	N	M	56	3	3	4	4	4	3
208	N	F	82	1	1	1	1	1	1
209	N	M	56	1	1	1	1	1	1
210	N	M	52	1	5	5	5	4	3
211	N	F	65	1	3	2	2	1	1
212	N	F	53	2	2	3	4	2	2
213	N	F	40	2	2	1	3	3	2
214	N	F	58	1	1	1	1	1	1
215	N	M	63	1	1	1	1	1	1
216	N	F	41	5	5	5	4	3	3
217	N	M	72	3	3	3	3	1	1
218	N	F	60	5	4	4	4	2	2
219	N	M	61	1	3	3	3	2	1

Table 3
Parameter estimates and standard errors (in parentheses) for the shoulder tip pain data with three association structures (positive parameter values indicate positive association)

Parameter	Independence	Exchangeable	Time-saturated
Treatment (active = 1, control = 2)	1.912 (0.488)	1.844 (0.492)	1.307 (0.457)
Sex (male = 1, female = 2)	−0.123 (0.488)	−0.122 (0.495)	−0.104 (0.474)
Age (years)	−0.026 (0.014)	−0.027 (0.013)	−0.023 (0.014)
Time (1–6)	−0.196 (0.074)	−0.193 (0.072)	−0.193 (0.059)

tures maintain their stability as long as there are at least two categories with large counts. The similarity of the exchangeable and independence models in these data is to be expected: There are the same number of observations for each subject and the same pattern of time-dependent structures maintain their stability as long as there are at least two categories with large counts. covariates, so the exchangeable model will not be superior to independence. It is most useful for clustered data, especially with varying numbers of observations in each cluster where none of the other models is appropriate and competing procedures like those of Stram, Wei, and Ware (1988) and Molenberghs and Lesaffre (1994) cannot be used.

Table 4
Odds ratios for the shoulder tip pain data, time-saturated model

	Time 2	Time 3	Time 4	Time 5	Time 6
Time 1	3.7	5.1	4.4	2.5	3.2
Time 2	—	23.2	15.5	6.2	4.2
Time 3		—	30.7	8.9	6.5
Time 4			—	20.5	15.2
Time 5				—	138.9

The conclusions drawn from this analysis are the same as those drawn from the independence model—that abdominal suction to remove the gas used to inflate the abdomen during surgery reduces the shoulder tip pain referred from the diaphragm. The original analysis also revealed a small effect on general abdominal pain and no effect on wound pain, as was expected from the biological hypothesis behind the study.

5. Conclusion

The common odds ratio model for association between ordinal responses allows GEE models with fewer association parameters than the alternatives and, thus, fills a gap between the independence model and the models of Lipsitz et al. (1994). This allows the use of GEE methods on sparser data sets and can indicate when greater efficiency will be possible by further modeling. The structure of the models, which allows more freedom in specifying correlations between times for given sets of categories than between categories for different sets of times, is often appropriate when there is a similar amount of information for all pairs of times but not for all pairs of categories. In relatively sparse data, it is especially important to structure the estimation of correlations between categories. Taking account of the ordinal structure in the data results in a more parsimonious model, but one that can still be expected to fit the data reasonably well. A common odds ratio model is consistent with any cumulative link model for the marginal probabilities, in contrast to a model based on a constant correlation matrix, which implies that the marginal distributions are the same for each group. This theoretical inconsistency would be expected to make estimation less efficient with correlation models, as the working correlation will be further from the true correlation. It is even possible to fit these odds ratio models to an ordinal variable that has been discretized differently at different time points. The regression parameters and odds ratio estimate the same population quantities irrespective of the actual division into categories, and it would only be necessary to allow the category-specific intercepts to be different at different times. While this may not be of great practical importance, correct behavior under amalgamation of categories is a property that should be possessed by models for ordinal data.

RÉSUMÉ

Les équations d'estimation généralisées (Liang, K. Y. et Zeger, S., 1986 *Biometrika* **73**, 13–22) permettent de modéliser des données longitudinales ou agrégées avec des hypothèses minimales sur leur structures de dépendance. Les structures d'association pour les données polytomiques requièrent généralement l'estimation d'une grand nombre de paramètres. Dans beaucoup d'applications mettant en jeu des données catégorielles, une structure ordinale est présente. Un ensemble de structures d'association et de méthodes de calcul pour les données ordinales catégorielles sont décrites, basées sur le odds ratio cumulatif, qui autorise des modèles beaucoup plus parcimonieux. Cela permet à la méthodologie des équations d'estimation généralisées d'être utilisées sur de plus petits jeux de données ordinales et avec moins d'effort consacré à la modélisation des associations. La méthode est illustrée par des jeux de données ordinales provenant d'études médicales.

REFERENCES

- Agresti, A. (1990). *Categorical Data Analysis*. New York: Wiley.
- Dale, J. R. (1986). Global cross-ratio models for bivariate, discrete, ordered responses. *Biometrics* **42**, 909–917.
- Ezzet, F. and Whitehead, J. (1991). A random effects model for ordinal responses from a crossover trial. *Statistics in Medicine* **10**(6), 901–907.
- Gange, S. J., Linton, K. L. P., Scott, A. J., DeMets, D. L., and Klein, R. (1995). A comparison of methods for correlated ordinal measures with ophthalmic applications. *Statistics in Medicine* **14**, 1961–1974.
- Heagarty, P. J. and Zeger, S. L. (1994). *Marginal regression models for clustered ordinal measurements*. Technical Report 790, Department of Biostatistics, Johns Hopkins University, Baltimore, Maryland.
- Jorgensen, J. O., Gilles, R. B., Hunt, D. R., Caplehorn, J. R. M., and Lumley, T. (1995). A simple and effective way to reduce postoperative pain after laparoscopic cholecystectomy. *Australian and New Zealand Journal of Surgery* **65**, 466–469.
- Koch, G. G., Landis, J. R., Freeman, J. L., Freeman, D. H., Jr, and Lehnen, R. J. (1977). A general methodology for the analysis of experiments with repeated measurement of categorical data. *Biometrics* **28**, 663–692.
- Liang, K. Y. and Zeger, S. (1986). Longitudinal data analysis using generalised linear models. *Biometrika* **73**, 13–22.
- Liang, K.-Y., Zeger, S. L., and Qaqish, B. (1992). Multivariate regression analyses for categorical data. *Journal of the Royal Statistical Society, Series B* **54**, 3–24.
- Lipsitz, S. R., Kim, K., and Zhao, L. (1994). Analysis of repeated categorical data using generalized estimating equations. *Statistics in Medicine* **13**(11), 1149–1164.
- Mardia, K. V. (1970). *Families of Bivariate Distributions*. London: Griffin.
- McCullagh, P. (1980). Regression models for ordinal data. *Journal of the Royal Statistical Society, Series B* **42**, 109–142 (with discussion).
- Miller, M. E., Davis, C. S., and Landis, J. R. (1994). The analysis of longitudinal polytomous data: generalized estimating equations and connections with weighted least squares. *Biometrics* **49**, 1033–1044.
- Molenberghs, G. and Lesaffre, E. (1994). Marginal modeling of correlated ordinal data using a multivariate Plackett distribution. *Journal of the American Statistical Association* **89**, 633–644.
- Plackett, R. L. (1965). A class of bivariate distributions. *Journal of the American Statistical Association* **60**, 516–522.
- Senn, S. (1993). A random effects model for ordinal responses from a crossover trial. *Statistics in Medicine* **12**, 2147–2151 (letter to the editor).
- Stram, D. O., Wei, L. J., and Ware, J. H. (1988). Analysis of repeated ordered categorical outcomes with possibly missing observations and time-dependent covariates. *Journal of the American Statistical Association* **83**, 631–637.

Received September 1994; revised March 1995; accepted July 1995.